AUTOOPS SCHOOL · COURSE STUDY SHEET

Linux Fundamentals for Beginners

Free course · 5 modules · 3 calendar months of access after Admin approval

About this course

Learn the fundamentals of Linux, essential terminal commands, file management, permissions, processes, and basic shell scripting. This course introduces the Linux skills needed to begin a DevOps learning journey.

What you will learn

- Understand Linux fundamentals
- Use basic Linux terminal commands
- Manage files and directories
- Understand users and permissions
- Learn basic process management
- Understand introductory shell scripting

Modules in this course

1. Introduction to Linux

   What Linux is, where it runs, and how to get a terminal to work in.

   Topics: What is Linux and why it dominates the server world, Linux distributions and choosing one, The terminal, shell and prompt explained, Installing a lab environment (VM, cloud, WSL)

2. Linux Commands and File Management

   Navigate, inspect, create and organise files and directories with confidence.

   Topics: Navigation: pwd, ls, cd and absolute vs relative paths, Creating, viewing and editing files: touch, cat, less, nano, Copy, move, rename and remove safely, Finding files: which, find, grep basics, Archives and compression: tar, zip, gzip

3. Users, Groups, and Permissions

   How Linux decides who may read, write or execute anything.

   Topics: Users, groups and the root account, Reading permission bits: rwx, numeric modes, chmod, chown and chgrp in practice, sudo and the principle of least privilege, Special permissions: setuid, setgid, sticky bit

4. Processes and System Administration

   Keep a Linux machine healthy: processes, services, storage and logs.

   Topics: Inspecting processes with ps and top/htop, Foreground, background and job control, Killing and prioritising processes, systemd services: start, stop, enable, status, Disk usage, mounts and file systems, Reading logs with journalctl and /var/log

5. Introduction to Shell Scripting

   Turn repeated commands into small, reliable automation scripts.

   Topics: Shebang, variables and quoting, Command arguments and exit codes, Conditionals and loops in bash, Writing your first automation script, Cron jobs and scheduling

How to use this sheet

Sign in at the AutoOps School student portal, open “Linux Fundamentals for Beginners”, and enroll. After Admin approval your 3-month access window starts and every PDF in the course materials section becomes available to view and download.